

Professional Summary

Motivated bioinformatics student with strong computational skills in -omics data analysis and gene network reconstruction. Experienced in RNA-seq analysis, variant calling pipelines, and network analysis for understanding regulatory mechanisms. Combines programming expertise in R, Python, and Java with molecular biology knowledge to investigate complex biological systems. Seeking to contribute to chromatin regulation research in neurodevelopmental disorders at the Ferguson Lab.

Education

2024–2028 **B.S. Biology w/ Specialization in Bioinformatics**, *University of California, San Diego*, **GPA: 3.8**
Second Year Student interested in genomics, chromatin regulation and neurodevelopmental research
Relevant Coursework:

- **BILD:** 1 (The Cell), 3 (Evolutionary Biology), 4 (Intro Biology Lab), 5 (R Biostatistics)
- **CHEM:** General Chemistry I-III (6A-C) with 7L Laboratory (Analytical/synthetic techniques)
- **MATH/CSE:** Calculus 1A-20B; CSE 8A-11 (Python/Java Programming), CSE 87 (Epidemiology)

Technical Skills

Programming Languages for Bioinformatics

- **R Programming:** ggplot2, Bioconductor, tidyverse, statistical analysis, network visualization
- **Python:** BioPython, Pandas, NumPy, scientific computing, -omics pipeline development
- **Java:** Core Java, OOP, data structures; Pipeline development: automated bash analysis
- **JavaScript/TypeScript:** Full-Stack, Node.js, ReactJS, API Integration, data visualization

-Omics Analysis & Sequencing

- **RNA-seq Analysis:** Expression quantification, differential analysis, multiome integration
- **Genomic Analysis:** BWA-MEM alignment, variant calling, SnpEff annotation, GraphAligner
- **Network Analysis:** Gene regulatory networks, pathway analysis, hierarchical clustering
- **Single-Cell Analysis:** 10x Genomics multiome (RNA+ATAC), pangenome reference alignment

Data Analysis & Visualization

- **Statistical Methods:** Hypothesis testing, regression modeling, PCA, hierarchical clustering
- **Data Processing:** Functional impact prediction, network/regulatory analysis, mutation spectrum
- **Visualization:** Interactive plots, heatmaps, network diagrams, multi-dimensional scaling
- **Linux/Unix:** Shell scripting, Docker, CLI tools, SSH, Oracle server admin, Nginx, Ubuntu VMs

Laboratory & Research Methods

- **Molecular Biology:** PCR, gel electrophoresis, DNA sequencing, cell culture techniques (agar)
- **Analytical Chemistry:** Gravimetric analysis, acid-base chemistry, spectrophotometry, titration
- **Data Analysis:** Hypothesis testing, regression modeling, statistical aggregation, p-values
- **Documentation:** Scientific writing, technical documentation, reproducible analysis reports, PRDs

Research Experience

2025 **Yeast Genome Regulatory Network Analysis**, *UC San Diego–Budin Lab*

- Developed comprehensive bioinformatics pipeline analyzing regulatory mechanisms in yeast model
- Implemented network analysis identifying gene-gene interactions and regulatory relationships

- Performed RNA-seq equivalent analyses through variant calling and annotation
- Discovered hierarchical conservation patterns in essential pathway regulation
- Performed mutation spectrum analysis and population structure studies on genomic scale
- Integrated genomic findings with biochemical data using network analysis approaches
- Technologies: Python (BioPython), R, Linux bash scripting, multiple bioinformatics tools
- Results: Identified distinct mutation patterns and discovered hierarchical conservation patterns, with corresponding regulatory impacts on sterol levels & lipid raft formation; *science.org* article.

2025 **UBIC Omics Odyssey - Multi-Omics Integration, 24-hour Competition**

- Analyzed 10x Genomics single-cell multiome PBMC dataset with paired RNA+ATAC reads
- Performed pangenome reference alignment and integrated multi-modal sequencing data
- Demonstrated ability to work with cutting-edge -omics technologies under time constraints

2024–Fall **BILD4 Epigenetics Research - DNA Hypomethylation Study, UC San Diego**

- Investigated epigenetic regulation through DNA hypomethylation patterns & chromatin modifications
- Studied ALDH2*rs671 variant's epigenomic impact on metabolic acetaldehyde pathways
- Demonstrated understanding of epigenetic mechanisms in development and disease processes

2023–HS **EpiGenAI - Epigenetic Analysis Portal, Epigenomics Research Project**

- Developed specialized portal for epigenetic data analysis focusing on DNA methylation patterns
- Integrated NLP for interpretation of methylation data in biological context with Embedding Models
- Built (now-deprecated) function-calling pipeline for LLMs on BigWig WGB-seq profiles (Epi-HK)

Professional Experience

2022 **Clinical Youth Volunteer, Cedars-Sinai Medical Center, Neurology Department, Los Angeles, CA**

- Supported neurological research and patient care in clinical neuroscience environment
- Directed communications between departments, gaining experience in professional collaboration
- Developed understanding of neurological documentation systems, research protocols, and data
- Demonstrated commitment to neuroscience through consistent and reliable weekly volunteer hours

Bioinformatics Projects

2025 **Genomic Data Analysis Pipeline Development**

- Created Python-based tools for processing large-scale genomic sequencing data
- Developed R visualization toolkit using ggplot2 and network analysis libraries
- Implemented automated pipelines for variant calling, annotation, and functional prediction

2024 **BILD5 Biostatistics - Neurological Data Analysis, UC San Diego**

- Analyzed simulated fMRI data measuring neuro-immune response patterns
- Applied R programming for statistical analysis of neurological experimental data
- Implemented linear regression models for brain activity correlation analysis

2024 **Data Integration Platform - First Web App**

- Developed Node.js platform integrating multiple data sources with API architecture
- Implemented data processing pipelines handling multi-modal inputs and schemas

Research Interests

- **Chromatin Regulation:** Histone modifications, chromatin dynamics, gene expression control
- **Neurodevelopmental/Computational Biology:** Gene regulatory networks in brain development
- **-Omics Integration:** Multi-modal data analysis (RNA-seq, ATAC-seq, proteomics) – From UBIC

Availability Available 3 hours/week with strong interest in long-term research commitment; lab certification.

New Experience

2025–Present **Ferguson Lab – Chromatin Regulation & Neurodevelopment, UC San Diego**

Investigating how BAP1 histone deubiquitinase regulates 3D genome architecture in brain development and its connection to neurodevelopmental disorders including autism, schizophrenia, & epilepsy.

Computational Pipeline Development

- Developed a Hi-C differential chromatin loop analysis pipeline using mariner/Bioconductor
- Implemented replicate-aware statistical framework with edgeR quasi-likelihood GLM (n=3)
- Built multi-resolution analysis system (5kb, 10kb, 25kb) with automated QC and visualization
- Created TAD/compartment differential analysis workflows integrating HOMER output
- Integrated ChIP-seq data (H3K27ac, H3K4me1) for enhancer-promoter loop classification
- Developed additional chromatin stripe differential analysis pipeline using Quagga with anchor-based quantification and tiered confidence scoring across P12-P60 developmental timepoints (BAP1 KO)

High-Performance Computing & Data Analysis

- Managed large-scale analysis on SDSC Expanse supercomputer (1.2 billion PE reads/sample)
- Ran upstream pipelines: HiC-Pro, Homer, nextflow for alignment, normalization, contact matrices
- Set up conda envs, MultiHiCompare with dchic and strawr for chromosome-specific contacts
- Technologies: R (Bioconductor, tidyverse), Python, bash, SLURM, HDF5, Singularity

Multi-Omics Integration

- Integrated Hi-C with CUT&RUN, ATAC-seq, and RNA-seq to link 3D chromatin with epigenetics
- Analyzed differential enhancer-promoter interactions across developmental timepoints (P12, P60)
- Experience with Taiji pipeline for ranking transcription factors based on ATAC-seq and RNA-seq
- Processed DNA methylation data via biomodal duet evoC pipeline (5mC/5hmC quantification)
- Ran modality XPLR Core Workflow for DMR calling, biological QC, methylation fraction analysis

Overall

- Contributing to manuscript linking BAP1 loss to genome-wide chromatin rearrangement